

# ACRME Calculation Logic Reference — All Scenarios (v2.2)

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## Revision v2.2 — 27 August 2026.

This document supersedes the previous calculation logic reference. Key changes in v2.2:

- **DR sizing formula replaced:** fixed `dr_ratio_*` percentage approach (v1) is superseded by the **max-not-sum distributed-portion formula** (Requirements v2.1 Appendix A.6 / Appendix D).
- **Reservation target floor formula added:** `Allocated VM Count + Configured Buffer` (CAP-003), distinct from the forecast-based growth formula.
- **Four new scenarios added:** DR destination requirement (max-not-sum), standby activation, deployment readiness gate, and customer seed record.
- **EU cross-geo DR region corrected:** Belgium Central → **Switzerland North** throughout.
- **Scenario 15** (fixed DR ratio parameters) retained for reference but marked **superseded**.

This document consolidates **every calculation logic** used by the Azure Capacity Reservation Management Engine (ACRME), organized by the scenario in which it fires. Each formula carries an evidence tag:

- `[Documented]` — backed by Microsoft Learn or Azure platform documentation.
- `[Decided]` — an approved design decision recorded in the ADR Decision Log.
- `[Derived]` — a logical consequence of the design; requires POC validation where noted.
- `[Assumed]` — a policy default or working hypothesis, tunable and not yet empirically validated.

**Sources:** Requirements Baseline v2.1/v2.2 (Appendix A-D); ADR-001 through ADR-004 v1.2; `acrme_production_readiness_review_and_architecture.md` (PRR §26–§32).

**Convention.** `vCPU` = `vCPU_per_instance` for the SKU family. `CVAL` (Customer Validation) and `NonProd` are interchangeable; scoring identifiers keep the `NonProd` spelling for engineering consistency. All raw ratios are clamped with `Clamp(x) = max(0, min(1, x))` before weighting.

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## Index of Scenarios

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| #  | Scenario                                                         | Core calculation                                                            | Status                           |
|----|------------------------------------------------------------------|-----------------------------------------------------------------------------|----------------------------------|
| 1  | Prod region derivation — customer picks a <b>geography</b>       | <code>argmax(PS_Prod)</code> over Standard regions                          | Current                          |
| 2  | Prod region validation — customer picks a <b>specific region</b> | HC-1..HC-10 gate + PS_Prod post-validation                                  | Current                          |
| 3  | Restricted region request                                        | Exception workflow (no scoring)                                             | Current                          |
| 4  | Middle East three-region deployment                              | <code>argmax(PS_Prod)</code> in-geo + <b>Switzerland North</b> cross-geo DR | <b>Updated</b>                   |
| 5  | CVAL / NonProd region selection                                  | <code>argmax(PS_NonProd)</code>                                             | Current                          |
| 6  | DR region selection                                              | <code>argmax(PS_DR)</code>                                                  | Current                          |
| 7  | Hard-constraint eligibility gate                                 | HC-3, HC-6, HC-7 arithmetic                                                 | Current                          |
| 8  | Quota group sizing                                               | Prod & NonProd+DR group budgets                                             | Current                          |
| 9  | DR floor accounting                                              | <code>DR_Floor_vCPU</code> , effective ceiling, headroom                    | <b>Updated</b>                   |
| 10 | Environment quota scoring                                        | <code>Quota_Score_{Prod,NonProd,DR}</code>                                  | Current                          |
| 11 | Capacity forecast / CR sizing                                    | <code>Forecast_Quantity</code>                                              | Current                          |
| 12 | Auto-increase trigger                                            | Utilisation thresholds + debounce                                           | Current                          |
| 13 | Emergency transfer & tier escalation                             | Tier 1/2/3, quota-neutral transfer                                          | Current                          |
| 14 | Scaling & API-budget model                                       | CRG cardinality, request-per-cycle                                          | Current                          |
| 15 | DR ratio parameters (fixed %)                                    | <code>dr_ratio_min/max/target</code>                                        | <b>Superseded by Scenario 17</b> |

| #  | Scenario                                  | Core calculation                                    | Status |
|----|-------------------------------------------|-----------------------------------------------------|--------|
| 16 | Reservation target & reconciliation floor | Allocated VM Count + Buffer (CAP-003)               | New    |
| 17 | DR destination requirement — max-not-sum  | MAX(source portions per destination) (A.6)          | New    |
| 18 | Standby activation (DR-019)               | associated → allocated staged acquisition           | New    |
| 19 | Deployment readiness gate                 | READY / QUOTA_DEFICIT / STALE_STATE / ... (RDY-002) | New    |
| 20 | Customer placement seed record            | First-placement persistence + reuse policy          | New    |

## Shared Building Blocks (used by several scenarios)

### Default scoring weights [Assumed]

```

alpha (α) = 0.30 # capacity / headroom signal
beta (β) = 0.20 # quota headroom signal
gamma (γ) = 0.25 # capacity-weighted distribution
delta (δ) = 0.15 # DR-coverage / overflow-integrity signal
epsilon(ε)= 0.10 # zone diversity
Clamp(x) = max(0, min(1, x)) # every component clamped to [0,1] before weighting

```

All five weights are tunable policy constants stored in `PlacementPolicy` (config-as-code, versioned). They sum to exactly 1.0 and apply across all three environment scoring formulas. [Assumed]

### Named regional-snapshot quantities [Decided]

```

nonprod_crg.effective_free = nonprod_crg.free_slots - dr_overflow_reserve
dr_crg.coverage_ratio      = dr_crg.quantity / potential_dr_demand
potential_dr_demand(region)= Σ prod_allocated for all customers whose dr_region = region

```

## Key formula disambiguation

| Formula                                                        | Purpose                                                    | Source       |
|----------------------------------------------------------------|------------------------------------------------------------|--------------|
| Target Reserved Capacity =<br>Allocated + Buffer               | Continuous reconciliation<br>floor (Scenario 16)           | CAP-003      |
| Forecast_Quantity =<br>ceil(Peak × (1+Growth) +<br>DR_Buffer)  | Proactive growth ahead of<br>demand (Scenario 11)          | ADR-004      |
| Destination DR Require-<br>ment(d) = MAX(source por-<br>tions) | DR standby sizing (Scenario<br>17)                         | A.6 / DR-017 |
| dr_ratio_* constants                                           | Superseded (Scenario 15) —<br>configuration reference only | PRR §32      |

## Scenario 1 — Prod Region Derivation (Customer Chooses a Geography)

**Trigger:** customer supplies a geography (e.g. “Europe”), not a specific region.

**Note (PLC-001/PLC-002):** Exact-region input (Scenario 2) is the preferred default. Geography-based selection is an exception path requiring approval and customer acknowledgement. [Decided]

**Logic:**

1. Build candidate set = **Standard Capacity Regions** in the chosen geography. Restricted regions are excluded before scoring. [Decided]
2. Score every candidate with `PS_Prod` from the versioned regional snapshot.
3. Select `derived Prod anchor = argmax(PS_Prod)`.
4. **Tie-break:** deterministic — first region in the Standard Capacity Region list order for that geography; also the cold-start default when no snapshot exists. [Derived]
5. Run sequential CVAL → DR selection (Scenarios 5 & 6) on the same Standard region pool.

## PS Prod(r) [Decided]

|                                                                |                   |
|----------------------------------------------------------------|-------------------|
| PS_Prod(r) =                                                   |                   |
| 0.30 × Clamp(nonprod_crg.effective_free / prod_crg.quantity)   | ← α: NonProd      |
| headroom signal                                                |                   |
| + 0.20 × Clamp(prod_crg.quota_headroom / prod_crg.quota_limit) | ← β: Prod quota   |
| headroom                                                       |                   |
| + 0.25 × Clamp(1 - prod_customer_count / total_customers)      | ← γ: distribution |
| fairness                                                       |                   |
| + 0.15 × Clamp(dr_crg.coverage_ratio)                          | ← δ: DR readiness |
| signal                                                         |                   |
| + 0.10 × Clamp(az_count / 3)                                   | ← ε: zone di-     |
| versity                                                        |                   |

`PS_Prod` is **dual-purpose**: derives the Prod region in Scenario 1 and is reused for post-selection validation in Scenario 2. Every candidate score and the policy version are written to the `OperationRecord` for deterministic replay. [Derived]

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## Scenario 2 — Prod Region Validation (Customer Supplies a Specific Region)

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**Trigger:** customer names an exact Azure region — the preferred/default input mode (PLC-001). [Decided]

**Logic:**

- **Standard Capacity Region:** validate against HC-1..HC-10 (Scenario 7). If eligible, it becomes the Prod anchor directly. `PS_Prod` computed for post-selection validation/audit only, not for selection.
- **Restricted Capacity Region:** do not score — route to the Exception Deployment Workflow (Scenario 3).

There is no `argmax` here. The customer's choice is authoritative once it passes the hard constraints. The first approved placement writes the **Customer Seed Record** (Scenario 20). [Decided]

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## Scenario 3 — Restricted Region Request (Exception Path)

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**Trigger:** the requested or only feasible region is a **Restricted Capacity Region**.

**Logic:** no scoring formula is evaluated. The request diverts to the manual Exception Deployment Workflow (governance approval, EC-1..EC-4, VR-1..VR-11, HC-9/HC-10 controls). Restricted regions never enter the scoring pipeline. [Decided]

**Exception conditions:**

| Code | Condition                                                          |
|------|--------------------------------------------------------------------|
| EC-1 | Customer requires the specific restricted region by contract       |
| EC-2 | No Standard region in the geography can supply sufficient capacity |
| EC-3 | Data-residency requirement binds to the restricted region          |
| EC-4 | Scenario 2 input — restricted regions can never be engine-derived  |

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## Scenario 4 — Middle East Three-Region Deployment (Updated — Switzerland North)

**Trigger:** a Middle East deployment requiring three environments.

**Logic:**

1. Candidate in-geo Standard regions = { Saudi Arabia Central, UAE North } (both Standard)
2. Score both with PS\_Prod.
3. Prod =  $\text{argmax}(\text{PS\_Prod})$  over the two.
4. CVAL = the remaining in-geo region (deterministic – only one candidate left).
5. DR = cross-geo extension region: Switzerland North (Europe)  
because no third in-geo Standard region exists to satisfy region separation.

**v2.2 correction (REG-002):** The cross-geo DR extension region was incorrectly cited as “Belgium Central” in earlier material. The authoritative placement configuration specifies **Switzerland North** as the cross-geo extension for Middle East. All references to Belgium Central are superseded by Switzerland North.

Middle East is subject to the `DR_NOT_OFFERED` policy flag (DR-014) where legal/data-sovereignty prevents an acceptable DR design. When `DR_NOT_OFFERED = true` for the geography, Step 5 above is bypassed and the seed record records `dr_region = NOT_OFFERED`. [Decided]

Cross-geo extension constraints from ADR-001 apply to the DR region (Switzerland North). [Decided]

## Scenario 5 — CVAL / NonProd Region Selection

**Trigger:** sequential step after the Prod anchor is fixed. Selects  $\text{argmax}(\text{PS\_NonProd})$  over eligible Standard regions.

**PS\_NonProd(r) — design-of-record** [Decided]

```
PS_NonProd(r) =
  0.30 × Clamp(nonprod_crg.effective_free / nonprod_crg.quantity)    ← α: effective NonProd headroom
+ 0.20 × Clamp(nonprod_crg.quota_headroom / nonprod_crg.quota_limit) ← β: NonProd quota headroom
+ 0.25 × Clamp(1 - nonprod_customer_count / total_customers)        ← γ: distribution fairness
+ 0.15 × Clamp(nonprod_crg.effective_free / nonprod_crg.quantity)    ← δ: overflow capacity health
+ 0.10 × Clamp(az_count / 3)                                          ← ε: zone diversity
```

**Corrected pilot variant** [Undocumented – architectural judgement]

The design-of-record above duplicates the same signal under  $\alpha$  and  $\delta$  (combined weight 0.45). The PRR’s

corrected pilot formula removes the duplication:

$$\text{PS\_NonProd} = 0.35 \times \text{Capacity} + 0.25 \times \text{Quota} + 0.25 \times \text{Distribution} \\ + 0.05 \times \text{DR\_Overflow\_Integrity} + 0.10 \times \text{Zones}$$

$$\text{Distribution} = 1 - \text{Region\_Assigned\_Demand} / \text{Total\_Assigned\_Demand}$$

Both variants are recorded. The design-of-record formula is the approved baseline; the pilot variant is the reviewer-recommended refinement. Neither is empirically validated yet.

**CVAL/DR co-location rule (PLC-010):** a customer's CVAL and DR may co-locate in the same destination

region. When co-located, CVAL capacity earmarked for DR activation must **not** be double-counted as both

live CVAL headroom and available DR headroom. The `CVALEarmarkRecord` tracks this. [Decided]

## Scenario 6 — DR Region Selection

**Trigger:** final sequential step. Selects `argmax(PS_DR)` over eligible Standard regions (or the cross-geo region for Middle East).

**PS\_DR(r)** [Decided]

$$\begin{aligned} \text{PS\_DR}(r) = & \\ & 0.30 \times \text{Clamp}(\text{dr\_crg.free\_slots} / \text{dr\_crg.quantity}) && \leftarrow \alpha: \text{DR CRG} \\ & \text{headroom} \\ & + 0.20 \times \text{Clamp}(\text{dr\_crg.quota\_headroom} / \text{dr\_crg.quota\_limit}) && \leftarrow \beta: \text{DR quota} \\ & \text{headroom} \\ & + 0.25 \times \text{Clamp}(1 - \text{dr\_customer\_count} / \text{total\_customers}) && \leftarrow \gamma: \text{distribution fairness} \\ & + 0.15 \times \min(1.0, \text{dr\_crg.coverage\_ratio} / \text{dr\_ratio\_target}) && \leftarrow \delta: \text{coverage-ratio health} \\ & + 0.10 \times \text{Clamp}(\text{az\_count} / 3) && \leftarrow \epsilon: \text{zone diversity} \end{aligned}$$

**Note on  $\delta$ :** `dr_ratio_target` in the PS\_DR formula is now a **configurable bootstrap reference** rather than a fixed 30–40% constant. See Scenario 17 for the authoritative DR sizing formula. [Decided]

## Scenario 7 — Hard-Constraint Eligibility Gate (Arithmetic Constraints)

Before any region is scored (Scenario 1) or accepted (Scenario 2), it must pass the hard constraints.

**HC-3 QUOTA\_FLOOR (per environment)** [Decided]

|          |                                     |        |                                        |                                  |
|----------|-------------------------------------|--------|----------------------------------------|----------------------------------|
| Prod:    | <code>Prod_Group_headroom(R)</code> | $\geq$ | <code>requested_vm_count</code> × vCPU |                                  |
| NonProd: | <code>nonprod_headroom(R)</code>    | $\geq$ | <code>requested_vm_count</code> × vCPU | # uses effective_nonprod_ceiling |
| DR:      | <code>dr_headroom(R)</code>         | $\geq$ | <code>target_dr_qty</code> × vCPU      |                                  |



## HC-6 DR\_COVERAGE\_FLOOR [Decided]

Region ineligible for DR if:  
 $(dr\_crg.free\_slots + nonprod\_crg.effective\_free) < customer\_requested\_dr\_slots$

## HC-7 DR\_FLOOR\_INTEGRITY [Decided]

REJECT NonProd placement in region R if:  
 $nonprod\_quota\_used(R) + (requested\_vm\_count \times vCPU) > effective\_nonprod\_ceiling(R)$

HC-3 checks raw group headroom; HC-7 additionally enforces floor compliance. Both must pass. On breach:

emit `DRFloorViolationDetected` (Critical severity).

## Minimum-headroom policy defaults [Decided]

```
min_prod_quota_headroom_vcpu    = 20
min_nonprod_quota_headroom_vcpu = 20
min_dr_headroom_vcpu             = 16
```

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## Scenario 8 — Quota Group Sizing (per Region)

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Every managed region has exactly two Azure Quota Groups. [Decided]

```
Prod_Group_Limit(region) =
  Prod_CRG_quantity × vCPU × (1 + prod_growth_buffer)
  prod_growth_buffer = 0.20

NonProd_DR_Group_Limit(region) =
  NonProd_CRG_quantity × vCPU × (1 + nonprod_growth_buffer)
  + DR_CRG_quantity    × vCPU
  + emergency_transfer_headroom_vcpu

nonprod_growth_buffer      = 0.20
emergency_transfer_headroom_vcpu = max_emergency_transfer_qty × vCPU
max_emergency_transfer_qty  = potential_dr_demand × emergency_transfer_pct
emergency_transfer_pct      = 0.30
```

The `emergency_transfer_headroom_vcpu` term is what makes Tier 2 DR expansion quota-neutral within the shared group. [Decided]

**Quota-as-governor principle (QUA-005):** quota allocation caps deployable capacity per product, environment, subscription, region, and VM family. Teams must justify quota increases — unallocated pooled quota is not automatically distributed. [Decided]

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## Scenario 9 — DR Floor Accounting (Engine-Enforced Sub-Limit) — Updated

Azure Quota Groups have no native intra-group sub-reservation; the engine enforces the DR floor by arithmetic. [Decided]

### Accounting formulas [Decided]

```
DR_Floor_vCPU(region)      = Destination_DR_Requirement(region) × vCPU
                             [v2.2: uses max-not-sum formula from Scenario 17]
                             [v1: was potential_dr_demand × vCPU × dr_ratio_max - su-
perseded]

Effective_NonProd_Ceiling = NonProd_DR_Group_Limit - DR_Floor_vCPU
NonProd_Headroom          = Effective_NonProd_Ceiling - NonProd_Used_vCPU
Group_Headroom            = Group_Limit - Group_Used
dr_crg.coverage_ratio      = dr_crg.quantity / Destination_DR_Requirement(region)
```

**v2.2 change:** `DR_Floor_vCPU` is no longer derived from `potential_dr_demand × dr_ratio_max`. It is now derived from `Destination_DR_Requirement(region)` — the max-not-sum value from Scenario 17.

This aligns the floor with the lean bootstrap model (DR-007) and the distributed DR topology (DR-016).

The fixed `dr_ratio_max = 0.40` constant is retained in Scenario 15 for legacy reference only.

### Dual-validation control [Decided]

A separate detector recomputes the DR floor from authoritative assignment data; any disagreement between

command-time and detector values disables automatic NonProd expansion until reconciled.

### Worked topology example [Decided]

```
Prod group      = 128 vCPU  (32 × D4s_v3, 4 vCPU each)
NonProdDR group = 80 vCPU
DR floor        = 32 vCPU  (= Destination_DR_Requirement for this region)
Effective NonProd ceiling = 48 vCPU  (80 - 32)
```

## Scenario 10 — Environment Quota Scoring (β Component Detail)

The generic `Quota_Score` is dispatched by environment type. [Decided]

```
Quota_Score_Prod(r)    = prod_group_headroom_vcpu / prod_group_limit_vcpu
Quota_Score_NonProd(r) = nonprod_headroom_vcpu   / effective_nonprod_ceiling_vcpu
Quota_Score_DR(r)      = dr_headroom_vcpu        / dr_floor_vcpu
```

Semantics differ by type: a DR score of `1.0` means maximum expansion room (DR CRG at zero), not maximum quota consumed. Zero-denominator guards apply. [Decided]

## Scenario 11 — Capacity Forecast / CR Sizing

Advisory forecast used to size CR quantity over a horizon — the **proactive growth path**, distinct from the continuous reconciliation floor (Scenario 16). [Decided]

```
Forecast_Quantity = ceil( Forecast_Peak × (1 + Growth_Buffer) + DR_Buffer )

Forecast_Peak      = predicted peak allocated-VM demand in the horizon      [De-
rived]
Growth_Buffer      = policy % for forecast uncertainty                      [As-
sumed]
DR_Buffer           = extra units required by approved recovery policy       [As-
sumed]
Forecast_Horizon ∈ { 30, 60, 90 } days                                     [Assumed]
```

**Lead-time alerting:** when forecast demand approaches 80% of quota limit, emit

`ForecastApproachingQuotaLimit` with 14-day lead time. [Documented]

Forecast recommendations remain advisory until model accuracy and false-positive rates are measured.

The horizon and buffer values are policy percentages stored in `PlacementPolicy`, not fixed constants.

## Scenario 12 — Auto-Increase Trigger (Steady-State Capacity Lifecycle)

When utilisation crosses a threshold, a `CapacityIncreaseRequest` is raised. [Decided]

```
Trigger auto-increase when utilisation ≥ threshold:
  dr_autoincrease_threshold      = 0.35
  prod_autoincrease_threshold    = 0.20
  nonprod_autoincrease_threshold = 0.20

Debounce cooldown = 30 min per (region + CRG type) after any trigger
```

Phase 1 requires operator approval. Auto-decrease excluded from Phase 1. The steady-state lifecycle runs **only** in `STEADY_STATE` engine mode (see ADR-003). [Decided]

### Normative 10-step steady-state capacity lifecycle

1. Detect threshold crossing.
2. Re-read current CR, quota, sharing, and assignment state.
3. Create `CapacityIncreaseRequest`.
4. Calculate target quantity ( `Forecast_Quantity` or `Allocated + Buffer` floor, whichever is higher).
5. Require operator approval (Phase 1).
6. Submit the quota action only if validated as required.
7. Wait for confirmed quota state — no assumed propagation SLA.
8. Update CR quantity.
9. Confirm the actual quantity.
10. Refresh the snapshot and close the request.

## Scenario 13 — Emergency Transfer & Tier Escalation (DR Event Active)

During an active DR event ( `DR_EVENT_ACTIVE` engine mode), emergency capacity is obtained by tier. [Decided]

|        |                      |                                                                                                         |
|--------|----------------------|---------------------------------------------------------------------------------------------------------|
| Tier 1 | DirectExpansion      | – additive DR expansion using existing headroom. Automated.                                             |
| Tier 2 | QuotaNeutralTransfer | – reduce NonProd CR, expand DR within the SAME quota group.<br>Policy-gated. Quota-neutral (see below). |
| Tier 3 | DestructiveTransfer  | – changes VM associations. Dual-approval + elevated RBAC.<br>BLOCKED in Phase 1.                        |

### Quota-neutral math for Tier 2 [Derived – P0C-31/32]

Because NonProd CR and DR CR draw from the SAME NonProd+DR quota group:  
 NonProd CR reduction → releases  $q \times \text{vCPU}$  to the GROUP pool  
 DR CR expansion → consumes  $q \times \text{vCPU}$  from the SAME GROUP pool  
 Net group headroom change  $\approx 0$  → no Azure quota-increase request needed  
 Tier RTO gated only by ARM operation time (minutes), not quota approval (hours)

$$\text{max\_emergency\_transfer\_qty} = \text{potential\_dr\_demand} \times \text{emergency\_transfer\_pct} \quad (\text{pct} = 0.30)$$

Escalation order: Tier 1 headroom available? → Tier 2 approved? → Tier 3 allowed? → manual .

## Scenario 14 — Scaling & API-Budget Model

Used to size the estate and stay within Azure Resource Manager throttling budgets. [Derived]

$$\begin{aligned} \text{CRG\_lower\_bound} &= 3 \times R && \# \text{ 3 CRGs (Prod/NonProd/DR) per region} \\ \text{CRG\_total} &= R \times \text{Eclass} \times Z \times \text{SKUset} \times \text{IsolationFactor} \end{aligned}$$

$$\begin{aligned} \text{Requests\_per\_cycle} &= \text{CRG\_reads} + \text{CR\_reads} + \text{sharing\_reads} + \text{quota\_reads} + \text{association\_reads} \\ \text{Average\_requests\_per\_second} &= \text{Requests\_per\_cycle} / 300 && \# \text{ 5-minute (300 s) cycle} \end{aligned}$$

### Documented ARM throttle baselines [Documented]

Compute RP read limit = 250 requests / 5 min / subscription  
 Compute RP write limit = 1,200 requests / hour / subscription

The adaptive throttle manager initialises with these baselines and adapts to observed 429/Retry-After .

## Reconciliation cadence [Assumed]

|                                 |                                                 |
|---------------------------------|-------------------------------------------------|
| Reconciliation loop target      | = 6 min (configurable; production interval TBD) |
| Critical targeted reconcile P95 | < 2 min                                         |
| Stable-resource state age P95   | < 15 min                                        |
| Drift detection                 | ≤ 2 reconciliation cycles                       |

## Scenario 15 — DR Ratio Parameters (Fixed %) — SUPERSEDED

**Status: Superseded by Scenario 17 (max-not-sum).** These constants are retained for configuration

reference and legacy comparison only. The fixed-percentage DR sizing model ( 30–40% of Prod ) was rejected in Requirements v2.0/v2.1 because it produces idle reserves estimated at **\$1.5M–\$5M/year**

at platform scale — directly contradicting the cost-reduction mandate. See Appendix D of the Requirements Baseline for the full derivation.

|                        |                |                                                                 |
|------------------------|----------------|-----------------------------------------------------------------|
| dr_ratio_min           | = 0.30         | [superseded – was HC-6 lower bound]                             |
| dr_ratio_max           | = 0.40         | [superseded – was DR_Floor_vCPU multiplier]                     |
| dr_ratio_target        | = [0.30, 0.40] | [superseded – now: configurable bootstrap per product/workload] |
| emergency_transfer_pct | = 0.30         | [still current – Scenario 13]                                   |
| prod_growth_buffer     | = 0.20         | [still current – Scenario 8]                                    |
| nonprod_growth_buffer  | = 0.20         | [still current – Scenario 8]                                    |

The **SUM** override (C-11 in the Configurable Items Register) remains available for specific customers or geographies that contractually require protection against concurrent regional failures. All other scopes use the max-not-sum formula (Scenario 17).

## Scenario 16 — Reservation Target & Reconciliation Floor (CAP-003) — New

**Trigger:** every reconciliation cycle compares current reserved quantity against this floor.

This is the **continuous floor formula** — distinct from the forecast-based growth formula (Scenario 11).

### A.1 Reservation target (CAP-003) [Decided]

Target Reserved Capacity = Allocated VM Count + Configured Buffer

- **Allocated VM Count** = VMs in **running** / **allocated** state consuming compute capacity.
- **Configured Buffer** = policy-defined headroom above allocated demand. Tunable per product/region/env/SKU. Not hard-coded.

- **Associated-but-deallocated VMs** do not force reservation retention (CAP-004). They are reported separately so teams understand restart risk, but do not add to the target.

## A.2 Reservation headroom

Reservation Headroom = Reserved Quantity - Allocated VM Count

## A.3 Reservation deficit

Reservation Deficit =  $\max(0, \text{Target Reserved Capacity} - \text{Reserved Quantity})$

## Reconciliation decision logic [Decided]

```
IF Reserved Quantity < Target Reserved Capacity:
    RAISE reservation (scale-up) toward target, subject to Azure availability.
    IF Azure cannot supply: hold current state, RAISE alert, expose buffer deficit.

IF Reserved Quantity > Target Reserved Capacity (consistently, after minimum-hold interval):
    LOWER reservation toward target – ONLY when all guards pass:
        - current_reserved > allocated_count           (never below allocated)
        - not within DR protection window
        - no active maintenance exclusion
        - cost policy permits reduction
```

## Zero-capacity and no-auto-delete policy (CAP-009/010) [Decided]

Reduce an unused managed reservation to ZERO; do NOT delete the reservation object. Normal reconciliation NEVER deletes CRGs or reservation definitions. Deletion uses a separate approved decommissioning workflow only.

# Scenario 17 — DR Destination Requirement — Max-Not-Sum (A.6/DR-017) — New

This scenario replaces the fixed `dr_ratio_*` sizing in Scenario 15.

## The single-failure assumption (DR-001) [Decided]

The programme plans for failure of **one** production region within a geography at a time. Simultaneous multi-region failure is outside the default guaranteed model.

## A.6 DR destination requirement — corrected formula [Decided]

```
Destination_DR_Requirement(d)
= MAX over each non-concurrent source region s protected by d (
    Workload_Portion(s → d)
)
```

Where:

- `d` = destination region hosting DR standby capacity
- `s` = source region whose production customers have DR in `d`
- `Workload_Portion(s → d)` = the quantity of VMs/vCPUs from source `s` that are assigned to land in destination `d` on failover (from the source→destination DR index, DR-018)

**Rationale:** because only one region fails at a time, destination `d` never simultaneously hosts failover from more than one source. It therefore needs standby capacity for its **largest** protected source only — never the sum of all sources. The standby slots are **shared / overcommitted** across mutually exclusive failure events. [Decided]

## A.7 DR capacity gap [Decided]

```
DR_Capacity_Gap(d) = max(0,
    Destination_DR_Requirement(d) - Usable_Destination_Capacity(d)
)
```

`Usable_Destination_Capacity(d)` may include approved bootstrap headroom, available CRG reservations, releasable CVAL capacity (after earmark check, Scenario 9/PLC-010), and capacity acquired through approved sharing or expansion.

## A.8 Shared-DR overcommit ratio (informational) [Derived]

```
Overcommit_Ratio(d) = SUM(Workload_Portion(s → d) for all s) / MAX(Workload_Portion(s → d))
```

A ratio > 1 quantifies: (a) the capacity and cost saved by max-not-sum sizing, and (b) the exposure if the single-failure assumption is ever violated. Required input for POC-011 risk sign-off.

## Observability requirement (OBS-002) [Decided]

Alert when a destination's usable capacity falls below `Destination_DR_Requirement(d)` — the max protected source — not below the sum. Dashboard must show per-destination max-source coverage (OBS-004).

## Conservative SUM override (C-11) [Decided]

```
Destination_DR_Requirement_Conservative(d) = SUM(Workload_Portion(s → d) for all s)
```

Used only where a customer or geography contract explicitly requires protection against concurrent regional failures. Configured per-scope in `PlacementPolicy`; no code change required.

## Worked example (from Requirements Appendix D) [Decided]

Region 2 holds DR standby for customers from Region 1 and Region 3:

| Source protected by R2   | Failover portion in R2 |
|--------------------------|------------------------|
| Region 1 (Cust1 + Cust5) | 120 cores              |
| Region 3 (Cust6)         | 80 cores               |

SUM sizing (old): Requirement(R2) = 120 + 80 = 200 cores ← over-provisions; super-seded

MAX sizing (new): Requirement(R2) = max(120, 80) = 120 cores

Saving at R2: 200 → 120 = 80 cores (40%) removed with no loss of protection.

Overcommit\_Ratio(R2) = 200 / 120 ≈ 1.67

→ R2's shared standby is 1.67× oversubscribed.

→ Platform-wide extrapolation: ~40–50% standby reduction ≈ \$0.6M–\$2.5M/year saved.

## Source→Destination DR Index entity (DR-018) [Decided]

The max-not-sum formula depends on an authoritative bidirectional mapping — a required state-store entity:

```
SourceDestinationDRIndex {
  source_region      : string
  destination_region : string
  customer_realm_id  : string
  standby_instance_set : list<InstanceRef>
  sku_family         : string
  quantity_vcpu      : int
  activation_state    : STANDBY | ACTIVATING | ACTIVE | FAILBACK_PENDING
  last_updated       : datetime
  policy_version      : string
}
```

This index is the reverse view of the customer seed record (Scenario 20). On a regional failure, the engine queries this index to determine exactly which standby sets to activate and where. [Decided]

## Scenario 18 — Standby Activation (DR-019) — New

**Trigger:** authorised DR declaration; engine mode transitions to `DR_EVENT_ACTIVE`.

### Per-customer activation workflow [Decided]

Each customer's DR instances transition `associated` → `allocated` (standby → active) in approved business-priority waves:

```
Priority Wave 0 (P0): Platform control planes and recovery orchestration infrastruc-
                    ture.
Priority Wave 1 (P-1): Highest-priority customer workloads.
Priority Wave N:      Remaining customers in business-priority order.
```

**DR-006 staged capacity acquisition per wave:**



Stage 1: Use approved bootstrap / pre-staged headroom (zero Azure wait time).  
 Stage 2: Allocate available reservation + quota already in the destination.  
 Stage 3: Shut down / disassociate eligible CVAL workloads (after DR-010 authorised trigger).  
 Stage 4: Share or reassign reservations within supported region/zone boundaries.  
 Stage 5: Allocate pooled quota to DR subscriptions.  
 Stage 6: Request additional Azure quota/capacity where required.  
 Stage 7: Report unrecoverable capacity gaps.

Stages 1-2 start immediately from bootstrap headroom — P0/P-1 customers begin recovery without waiting

on Azure deallocation/disassociation APIs (which are throttled during regional events). [Decided]

## Activation state tracking [Decided]

```
ActivationRecord {
  customer_realm_id : string
  source_region     : string
  destination_region : string
  priority_wave      : int
  activation_state   : PENDING | ACTIVATING | ACTIVE | FAILED | FAILBACK_PENDING
  stage_reached      : int          # 1-7 per DR-006
  capacity_gap_vcpu  : int          # 0 if fully provisioned
  activated_at       : datetime
  approved_by        : string
  incident_id        : string
}
```

Activation state is tracked per customer, auditable, and reversible on failback (DR-013). [Decided]

## Failback reversal [Decided]

On failback: ACTIVE → FAILBACK\_PENDING → STANDBY in reverse priority order. The seed record and DR index are preserved; no re-derivation of placement. History, audit trail, and DR capacity state are retained through the full failback cycle.

---

## Scenario 19 — Deployment Readiness Gate (RDY-002) — New

---

**Trigger:** every AEP-triggered deployment before the first environment is provisioned.

## Machine-readable readiness states [Decided]

|                      |                                                                                       |
|----------------------|---------------------------------------------------------------------------------------|
| READY                | All gates pass; deployment may proceed.                                               |
| READY_WITH_RISK      | All gates pass but one or more advisory risks detected (e.g. buffer below target).    |
| QUOTA_DEFICIT        | Required deployment quota > available quota in the deploying subscription.            |
| RESERVATION_DEFICIT  | Reservation exists but quantity insufficient; approved over-allocation not in effect. |
| CAPACITY_UNAVAILABLE | Azure cannot supply the requested SKU/zone at this time.                              |
| STALE_STATE          | Capacity/quota snapshot exceeds max-age threshold; cannot drive a safe placement.     |
| POLICY_BLOCKED       | A governance policy (hard constraint, exception, approval) blocks the deployment.     |
| VALIDATION_REQUIRED  | One or more validation steps (e.g., POC-gated behaviours) not yet confirmed.          |

## Readiness gate logic (RDY-001) [Decided]

All of the following must pass for `READY` :

1. `target_region`  $\in$  `approved_region_catalogue` (REG-001)
2. `az_count`  $\geq 1$  in `target_region` (CAP-011)
3. `sku`  $\in$  `supported_skus(target_region)`
4. `reservation_policy` known for (`subscription`, `region`, `zone`, `sku`)
5. `reservation` exists (if required by policy)
6. `reserved_quantity`  $\geq$  `requested_count` OR `approved_over_allocation` active
7. `consumer_subscription_quota`  $\geq$  `requested_units` (QUA-013 – POC-gated)
8. `snapshot_age`  $\leq$  `max_snapshot_age_seconds` (RDY-004)
9. no blocking policy exception or hard constraint violation

## Staleness check (RDY-004) [Decided]

```
IF now() - snapshot.collected_at > max_snapshot_age:
    RETURN STALE_STATE
    ACTION: refresh synchronously OR block with STALE_STATE – never drive placement
    from stale data
```

## Quota deficit formula (A.5) [Decided]

```
Quota_Deficit = max(0, Requested_Deployment_Units - Available_Quota)
Available_Quota = Assigned_Regional_VM_Family_Quota - Current_Regional_VM_Family_Usage (A.4)
```

## Scenario 20 — Customer Placement Seed Record (PLC-003/004/005) — New

**Trigger:** first approved production-region placement for a customer in a geography.

## Seed record creation [Decided]

```
CustomerSeedRecord {
  customer_realm_id      : string      # authoritative customer/realm identifier
  geography              : string      # e.g. "NorthAmerica", "Europe"
  production_region      : string      # exact Azure region name
  cval_region            : string      # exact Azure region name
  dr_region              : string | "NOT_OFFERED"
  products_covered       : list<string> # all products seeded by this record
  decision_timestamp     : datetime
  policy_version         : string      # PlacementPolicy version used
  capacity_snapshot_ref  : string      # snapshot ID that drove the decision
  exception_ref          : string | null # approval ID if Restricted region or geo-
exception
  approval_metadata     : ApprovalRecord
  input_mode             : "SPECIFIC_REGION" | "GEOGRAPHY_EXCEPTION"
}
```

## Seed reuse policy (PLC-004) [Decided]

```
IF CustomerSeedRecord exists for (customer_realm_id, geography):
  READ seed record – do NOT re-invoke the placement engine.
  Subsequent products and environments use the seeded production/CVAL/DR regions.
  Engine is NOT re-run per product once seeded.
```

## Controlled seed change (PLC-005) [Decided]

```
IF seed change requested:
  REQUIRE approved migration/exception workflow.
  REQUIRE impact analysis across all products covered by the seed.
  PROHIBIT automatic regeneration on: upgrades, rebuilds, routine deployments.
```

---

## A. Core Formula Reference (Appendix A — Requirements v2.1)

| Label | Formula                                                                            | Source  |
|-------|------------------------------------------------------------------------------------|---------|
| A.1   | Target Reserved Capacity =<br>Allocated VM Count + Con-<br>figured Buffer          | CAP-003 |
| A.2   | Reservation Headroom = Re-<br>served Quantity - Allocated<br>VM Count              | CAP-003 |
| A.3   | Reservation Deficit =<br>max(0, Target Reserved Ca-<br>pacity - Reserved Quantity) | CAP-003 |
| A.4   | Available Quota = Assigned<br>Regional VM-Family Quota -<br>Current Usage          | QUA-002 |
| A.5   | Quota Deficit = max(0, Re-<br>quested Deployment Units -<br>Available Quota)       | QUA-007 |
| A.6   | Destina-<br>tion_DR_Requirement(d) =<br>MAX over s (Work-<br>load_Portion(s → d))  | DR-017  |
| A.7   | DR_Capacity_Gap(d) = max(0,<br>DR_Requirement(d) - Us-<br>able_Capacity(d))        | DR-017  |
| A.8   | Overcommit_Ratio(d) =<br>SUM(portions) /<br>MAX(portions)                          | DR-017  |

## **B. Consolidated Policy-Constant Table (v2.2)**

---

| Constant                               | Value                                          | Used in Scenario(s) | Status                |
|----------------------------------------|------------------------------------------------|---------------------|-----------------------|
| alpha / beta / gamma / delta / epsilon | 0.30 / 0.20 / 0.25 / 0.15 / 0.10               | 1, 5, 6             | Current               |
| prod_growth_buffer                     | 0.20                                           | 8                   | Current               |
| non-prod_growth_buffer                 | 0.20                                           | 8                   | Current               |
| emergency_transfer_pct                 | 0.30                                           | 8, 13               | Current               |
| min_prod_quota_headroom_vcpu           | 20                                             | 7                   | Current               |
| min_nonprod_quota_headroom_vcpu        | 20                                             | 7                   | Current               |
| min_dr_headroom_vcpu                   | 16                                             | 7                   | Current               |
| dr_autoincrease_threshold              | 0.35                                           | 12                  | Current               |
| prod/non-prod_autoincrease_threshold   | 0.20                                           | 12                  | Current               |
| Debounce cooldown                      | 30 min                                         | 12                  | Current               |
| Reconciliation loop target             | 6 min (configurable)                           | 14                  | Current               |
| ARM read / write baseline              | 250 per 5 min / 1,200 per hour                 | 14                  | Current               |
| Forecast_Horizon                       | 30 / 60 / 90 days                              | 11                  | Current               |
| dr_ratio_min                           | 0.30                                           | 15                  | <b>Superseded</b>     |
| dr_ratio_max                           | 0.40                                           | 15                  | <b>Superseded</b>     |
| dr_ratio_target                        | [0.30, 0.40]                                   | 15                  | <b>Superseded</b>     |
| DR bootstrap target                    | Configurable per product/workload — no fixed % | 17                  | <b>Replaces ratio</b> |

| Constant                         | Value                                   | Used in Scenario(s) | Status                               |
|----------------------------------|-----------------------------------------|---------------------|--------------------------------------|
| Max-not-sum default              | MAX(source portions)                    | 17                  | Current                              |
| SUM override (C-11)              | SUM(source portions) — per-scope opt-in | 17                  | Current                              |
| EU cross-geo DR extension region | <b>Switzerland North</b>                | 4                   | <b>Updated (was Belgium Central)</b> |

## C. Evidence & Maturity Summary

| Formula                                  | Evidence level               | Validation required                                    |
|------------------------------------------|------------------------------|--------------------------------------------------------|
| PS_Prod , PS_NonProd , PS_DR (scoring)   | [Decided] — design-of-record | Shadow/recommendation mode until empirically validated |
| HC-3, HC-6, HC-7 (hard constraints)      | [Decided]                    | Ready for Phase 1                                      |
| DR floor accounting (Scenario 9)         | [Decided]                    | Updated to use max-not-sum input                       |
| Tier 2 quota-neutral math (Scenario 13)  | [Derived]                    | POC-31/POC-32 required                                 |
| Max-not-sum DR sizing (Scenario 17)      | [Decided]                    | POC-011 required before production dependency          |
| Consumer-subscription quota (QUA-013)    | [Assumed]                    | POC-001 — <b>top technical unknown</b>                 |
| Reservation target floor (Scenario 16)   | [Decided]                    | Ready for Phase 1                                      |
| Standby activation staging (Scenario 18) | [Decided]                    | Dependent on POC-005 (VM state semantics)              |
| Deployment readiness gate (Scenario 19)  | [Decided]                    | Ready for Phase 1                                      |
| Customer seed record (Scenario 20)       | [Decided]                    | Ready for Phase 3                                      |

All constants are policy defaults stored in `PlacementPolicy` (config-as-code, versioned). Tuning any constant requires updating the config; no code change is needed. The `dr_ratio_*` constants are retained in the codebase as fallback references for the SUM override (C-11) only.

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Document version 2.2 — 27 August 2026. Next review: upon POC-001 / POC-011 results.